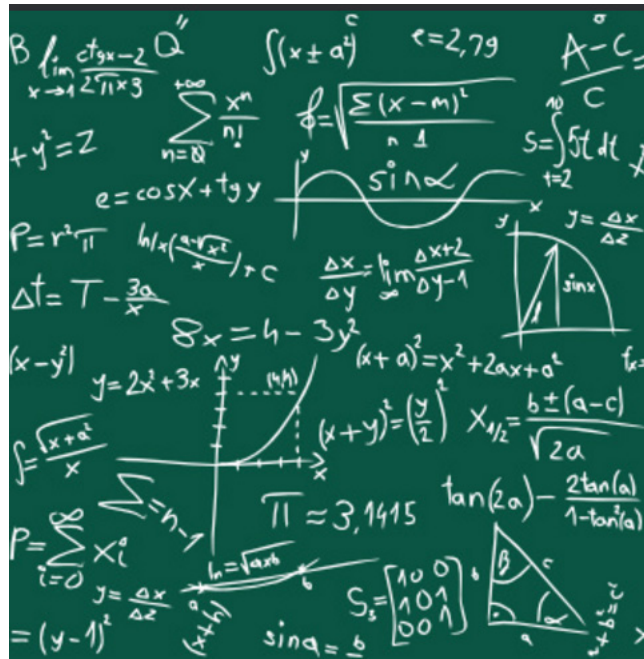


Composite functions



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

Contents

1 Composite functions

2

1 Composite functions

Definition 1: of composite function

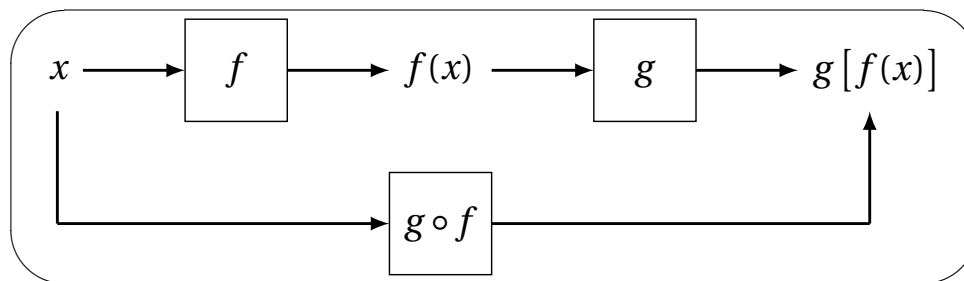
Given two functions:

$$f: E \rightarrow \mathbb{R} \quad \text{and} \quad g: F \rightarrow \mathbb{R},$$

if $f(E) \subseteq F$ (i.e., if for every $x \in E$ we have $f(x) \in F$), one can define the function $h: E \rightarrow \mathbb{R}$, the **composite** of f and g (in this order), denoted by the symbol $g \circ f$, through the formula

$$h(x) = (g \circ f)(x) = g[f(x)]$$

- The composition scheme is the following:

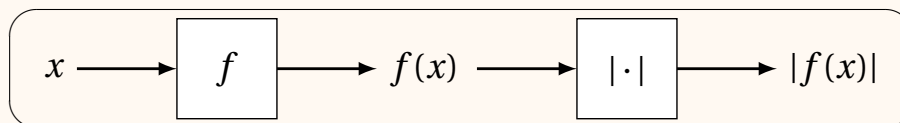


Example 1: of composite function

The function $x \mapsto |f(x)|$ is actually “composed” of two functions:

- given x , we compute $f(x)$
- once $f(x)$ is computed, we compute $|f(x)|$

This amounts to operating in series with two black boxes, the first corresponding to f , the second to its absolute value, according to the following scheme:



- It may happen that both compositions $(g \circ f)$ and $(f \circ g)$ are well defined, but in general

$$(f \circ g) \neq (g \circ f)$$

In other words, **the commutative property does not hold.**

Example 2: of composition of functions

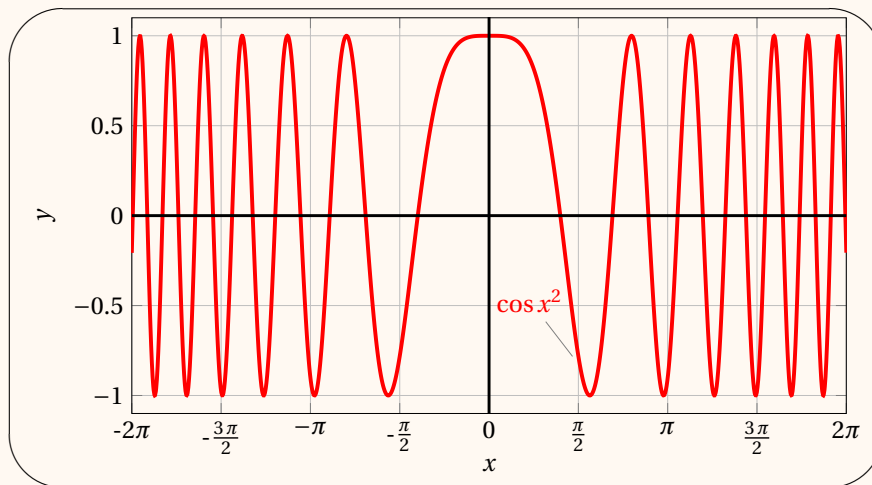
Consider the functions:

$$f: \mathbb{R} \rightarrow \mathbb{R}, \quad f: x \mapsto x^2 \quad \text{and} \quad g: \mathbb{R} \rightarrow \mathbb{R}, \quad g: x \mapsto \cos x$$

that is, $f(x) = x^2$ and $g(x) = \cos x$.

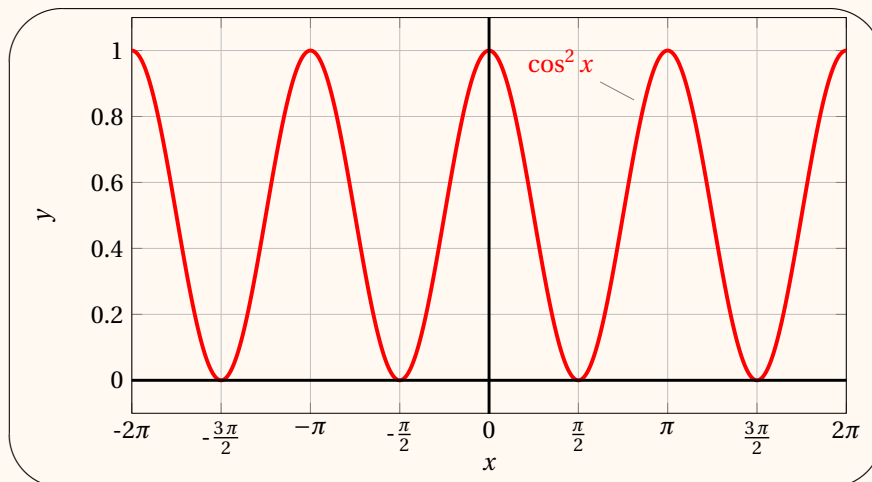
1. Since g is defined on all of $\mathbb{R}$, $h = g \circ f$ is well defined on $\mathbb{R}$ and the following formula holds

$$h(x) = (g \circ f)(x) = g[f(x)] = \cos x^2$$



2. Since f is defined on all of $\mathbb{R}$, $k = f \circ g$ is well defined on $\mathbb{R}$ and the following formula holds

$$k(x) = (f \circ g)(x) = f[g(x)] = \cos^2 x$$



Example 3: of composition of functions

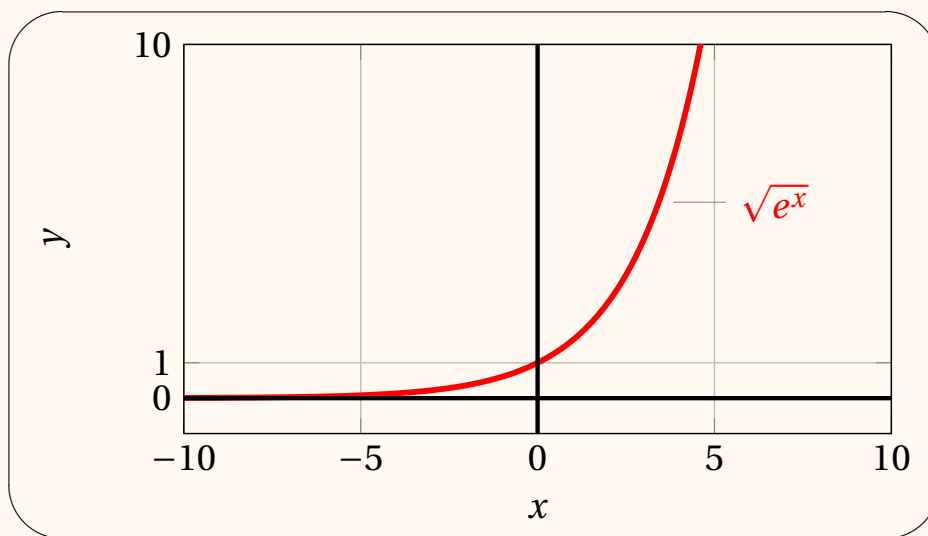
Consider the functions:

$$f: \mathbb{R} \rightarrow \mathbb{R}, \quad f: x \mapsto e^x \quad g: [0, +\infty) \rightarrow \mathbb{R}, \quad g: x \mapsto \sqrt{x}$$

that is, $f(x) = e^x$ and $g(x) = \sqrt{x}$.

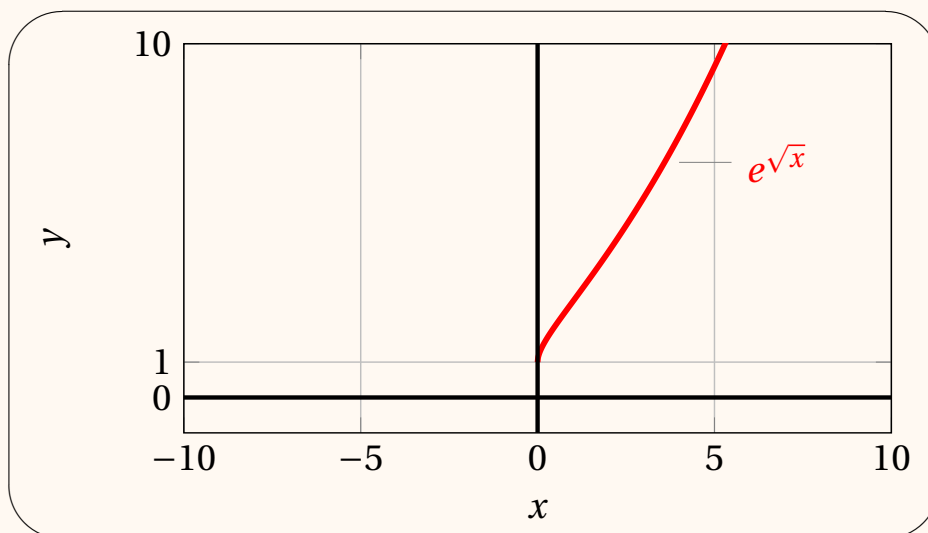
1. Since g is defined on $[0, +\infty)$ and $f(x) > 0$ for every $x \in \mathbb{R}$, $h = g \circ f$ is well defined on $\mathbb{R}$ and the following formula holds

$$h(x) = (g \circ f)(x) = g[f(x)] = \sqrt{e^x}$$



2. Since g is defined only on $[0, +\infty)$, $k = f \circ g$ is well defined only on $\mathbb{R}_+$ and the following formula holds

$$k(x) = (f \circ g)(x) = f[g(x)] = e^{\sqrt{x}}$$



- The composition operation can be extended to three or more factors. One can verify that if the composition $(f \circ g) \circ r$ exists, then $f \circ (g \circ r)$ also exists and they are equal

$$(f \circ g) \circ r = f \circ (g \circ r)$$

In other words, **the associative property holds**.

- If a function $f : D \rightarrow \mathbb{R}$ is such that $f(D) \subseteq D$, then it can be composed with itself

$$f^2 = f \circ f \text{ that is } f^2(x) = f[f(x)]$$

f^2 is called the **second iterate** of f . Similarly, f^n is called the **n -th iterate function** of f .